

Parkflow

Parking Slot Booking and Management System

A Project Synopsis Report Submitted To



Where Success is a Tradition

SAGE UNIVERSITY, INDORE

**Bachelor of Technology
in
Computer Science & Technology**

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Title - PARKING SLOT BOOKING AND MANAGEMENT SYSTEM

Problem Statement

Parking management is a common challenge in urban areas due to the increasing number of vehicles and limited parking spaces. Users often face difficulties in finding available parking slots, while parking operators rely on manual processes for managing reservations and occupancy. The lack of real-time parking information results in inefficient space utilization and inconvenience for both users and operators. Therefore, a digital parking management system is required to simplify parking reservations and improve parking facility management.

Objectives and Scope of the Project

Objectives

- Provide real-time parking slot availability.
- Enable advance slot booking.
- Support secure online payment.
- Generate QR Code/PIN-based booking confirmation.
- Improve parking facility management.
- Maintain booking and transaction records.
- Simplify parking facility management.

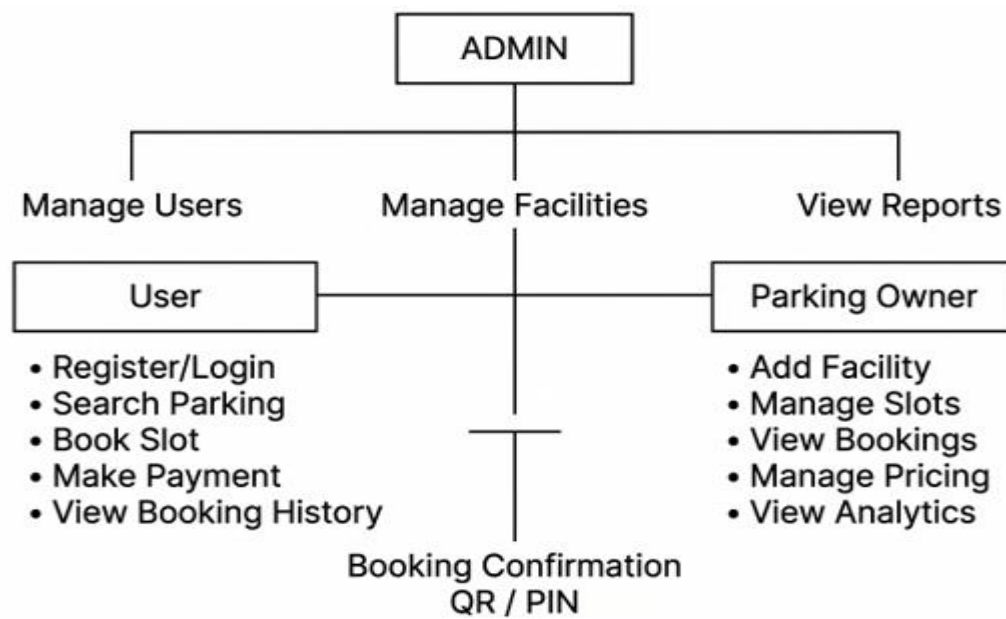
Scope

The system is designed for shopping malls, hospitals, airports, railway stations, educational institutions, and commercial complexes. It allows users to search, reserve, and pay for parking slots online, while parking owners can manage facilities, slots, and reservations through a centralized platform.

Process Description

The system allows users to search for parking facilities, view available slots, reserve a slot, make payments, and receive booking confirmation. Parking owners can manage facilities, slots, pricing, and booking records. Administrators can monitor users and parking facilities.

Use Case



Methodology

The project follows the Agile Development Methodology. Development is carried out in iterative sprints covering requirement analysis, system design, frontend and backend development, database integration, testing, and deployment. Continuous testing and feedback are incorporated throughout the development process.

Resources and Limitations

Hardware Requirements

- Intel Core i3 or higher
- 4 GB RAM
- 50 GB Storage
- Internet Connection

Software Requirements

- HTML5, CSS3, JavaScript

- Node.js, Express.js
- MySQL
- Visual Studio Code , Git & GitHub
- Google Maps API
- Razorpay Payment Gateway

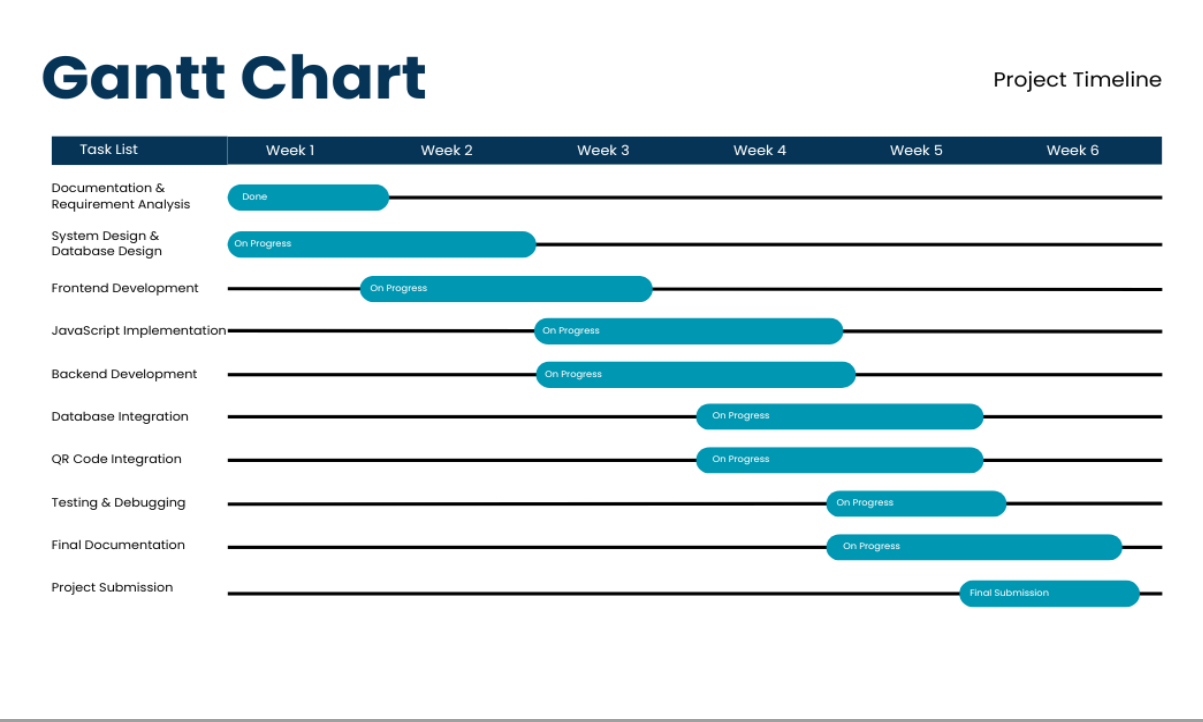
Limitations

- Requires internet connectivity.
 - Accuracy depends on regular slot updates.
 - Initial version supports web platform only.
 - Payment gateway transaction charges may apply.
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Future Scope

- Dynamic Pricing Based on Demand
- Mobile Application Development
- Multi-City Parking Integration
- EV Charging Slot Reservation
- Subscription-Based Parking Passes
- Real-Time Occupancy Analytics
- Vehicle Entry and Exit Tracking
- Loyalty and Reward Programs

Project Schedule (Gantt Chart)



Conclusion

The Parking Slot Booking and Management System provides a digital solution for managing parking facilities efficiently. It enables users to reserve parking spaces online and helps parking operators manage reservations, occupancy, and payments effectively. The system improves parking utilization and enhances user convenience through real-time booking and digital verification.

References

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3. Express.js Documentation.
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5. Google Maps Platform Documentation.
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7. MDN Web Documentation.